

ALL ABOUT HOME SOLAR SYSTEMS

Know the right current controller, inverter and battery combination to get the most out of your PV solar panel



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Going for home solar power is a wise decision as solar energy is not only natural but is free to us unlike any other source of electrical power. So you will be helping to reduce greenhouse gas emissions. However, home solar power systems need significant initial investment. So you have to be careful.

Unfortunately, for unscrupulous companies, consumer ignorance about photovoltaic (PV) solar power is often an opportunity to cheat. In fact, many vendors are in the solar business to chase your money with little interest in the technology, environmental benefits or your needs. Here is some information that will help you fortify yourself against being shortchanged.

PV solar modules convert energy from the sun into free and non-polluting DC power. Unfortunately, this power cannot be directly used due to varying light intensity of the sun over the day. You need to use a current controller to convert it into a constant voltage under all conditions of light.

The only controller type that you have to buy is the one that harvests the maximum solar energy from the sun. This is called maximum peak power tracking (MPPT) controller. MPPT helps you to harvest about 30 per cent more energy from the solar module than any other type. Solar panels are expensive and the MPPT controller costs maximum 10 per cent of the cost of the panel. Since only an MPPT controller gives 30 per cent more energy from the same panel, this investment

is fully justified.

The sun shines bright during the day. In India, often there is no electricity even during the day for many hours. During daytime, the MPPT controller can directly feed DC into your inverter and you can use that AC power for feeding electric fans, television or computers. For this, you need only a small-capacity battery. The inverter directly uses the solar panel output to give you AC supply. If you are using it for office, you need power mostly during the day, except for some time in the early morning and late evening. A small-size battery is good enough for this.

However, home users need solar power at night after the sun has set. For this, one needs to accumulate all the available solar energy during the day and then use it at night. In such a case, you need a battery with big enough capacity to accumulate all the available solar energy from the sun and deliver it to the inverter at night feeding your 230V AC loads like lights, fans, PCs and TVs.

Depending on your energy need at night, your supplier will recommend certain battery capacity; its voltage and ampere-hour (Ah) capacity. Small home inverters of 300-1500W rating usually work off one or two 12V lead-acid batteries depending on the inverter power and its input voltage rating.

Electrical power and energy often confuse people. But it is easy to understand the difference between the two. Your CFL light, normal bulb, fan and TV are rated by their power consumption and the unit used is watt (W) or volt-ampere (VA). You can add all these to find out how much power you